David Amezcua Santiago

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Education

Iowa State University Anticipated May 2027
Bachelor of Science, Computer Engineering, GPA: 3.3
Des Moines Area Community College August 2022 – May 2024
Associate of Science in Engineering, GPA: 3.3

Technical Skills

- **Programming Languages:** C, C++, Python, Java
- **Hardware / Architecture:** RISC-V, VHDL, Verilog, FPGA
- **Developer Tools / Technologies:** Linux, Git, Android Studio, Postman, LTSpice, Questa Sim, Quartus Prime
- **Spoken Languages:** English, Spanish

Experience

IINSPIRE LSAMP Undergraduate Research Assistant
Advisor: Anwesha Sarkar June 2024 – August 2024
Department of Electrical Engineering and Computer Engineering, Iowa State University

- NSF-funded research on atomic force microscopy (AFM) image sets using vision language models (VLMs)
- Applied text and image based multimodal models to atomic force microscopy (AFM) image sets to produce faster and more consistent image analysis
- Presented at IINSPIRE LSAMP and Iowa State University Research Symposiums

Projects

LLM Agentic System for Circuit Card Design August 2026 – Present
Senior Design Project – Computer Engineering

- Developing an agentic LLM system to automate early-stage circuit card design processes, including design trade studies, component selection, and hardware design exploration
- Building within a rootless Docker environment on a cloud-based server using Python, Linux, and locally hosted LLMs, with an emphasis on offline execution, traceable agent actions, verification, and model interchangeability
- Exploring MCP, GraphRAG, and retrieval technologies to support multi-agent workflows, engineering knowledge retrieval, and generation of circuit design artifacts

Flextasy – Android Fitness Application January 2026 – May 2026
Front-End Developer

- Designed and developed an Android fitness application featuring personalized six-month fitness plans based on user goals, activity levels, and body metrics
- Implemented user-facing features for workout and meal planning, daily nutrition and exercise tracking, challenges, leaderboards, social connections, and direct/group messaging
- Developed interfaces supporting personal, fitness coach, and dietitian accounts, enabling professionals to create, assign, update, and manage customized workout and meal plans for clients

Make to Innovate (M:2:I) August 2024 – January 2025
MAVRIC Electrical Systems Team

- Developed an analysis report to enhance the performance of the rover's electrical systems
- Collaborated on engineering design to identify areas of improvement
- Engaged in team brainstorming sessions to develop innovative solutions for optimizing rover functionality